Deepak Neupane

Software Engineer — Embedded Build Systems · CI/CD · Embedded Security
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Summary

Software engineer with six years at HL Mechatronics owning complex engineering workflows end to end: the build orchestration, CI pipelines, and internal tools a 40+ engineer team uses to build, test, and release ASIL D steering software on Infineon AURIX TriCore (ISO 26262, MISRA C). Works directly with the engineers who use the software, from requirements through production, and debugs ECU issues alongside plant engineers when the line is down.

Experience

Software Engineer III — HL Mechatronics · 2020 – Present

Build orchestration and CI/CD

- Wrote the Python and Groovy build orchestration that runs on Jenkins and GitLab for 10+ production and advanced-development programs. 40+ software, architecture, and motor controls engineers build, test, and ship 3+ customer releases a week through it. Primary escalation point when a build or pipeline breaks for any projects.
- Cut MCU diagnostics fault-injection testing from 3+ days to under 4 hours per release by designing a multi-stage CI workflow that chains code instrumentation, test case generation, and on-target execution, with logs and results at every stage. It exercises the ASIL D safety mechanisms and has caught performance and communication defects before customer release.
- Own configuration management for platform software: repository structure, branching strategy, build servers, and release packages in GitLab and PTC Integrity.

Developer tools

- Built an internal developer tools portal in Django, in production: UDS security access key computation for development ECUs and CMAC calculation over firmware binaries.
- Worked with the prototype team to build a build-request tracker in Django, now in production. Engineers register prototype parts and submit hardware modification requests that the prototype team picks up and works.
- Set up and maintain Vector CANoe, CANalyzer, and CANape workspaces for 10+ projects, used by 40+ engineers to talk to ECUs.

AUTOSAR, security, and diagnostics

- Configure and integrate AUTOSAR basic software (BSW) modules from Vector and EB tresos stacks into platform software. Configured and verified CMAC-based secure boot across the bootloader and HSM on AURIX TriCore, and ran penetration tests against the ECU.
- Wrote the in-house Python libraries behind our tracetronic ECU-TEST integration tests for UDS diagnostics and XCP calibration, run on bench hardware with a live motor and traced to requirements in PTC Integrity. Trained junior engineers to write and review tests with them.
- Work on the bench with plant production engineers to root-cause UDS and CAN issues that stop the production line, under time pressure.

Education

- **M.S., Software Engineering and Information Technology** — Oakland University, Rochester, MI · 2024

Technical Skills

- **Languages:** Python, C, Groovy, Java, C#; Django
- **Build & CI/CD:** Jenkins, GitLab CI, Git, PTC Integrity (configuration management, requirements traceability), Linux and Windows build servers
- **Embedded & security:** Infineon AURIX TriCore, AUTOSAR BSW (Vector, EB tresos), secure boot, HSM, CMAC, Tasking and Wind River compilers
- **Test & tools:** tracetronic ECU-TEST, fault injection, penetration testing, Polyspace, Vector CANoe/CANape/DaVinci Configurator, iSYSTEM winIDEA/BlueBox
- **Standards:** ISO 26262 (ASIL D), MISRA C, AUTOSAR Classic, UDS (ISO 14229), XCP, CAN